



The Ultimate Trig Cheat Sheet



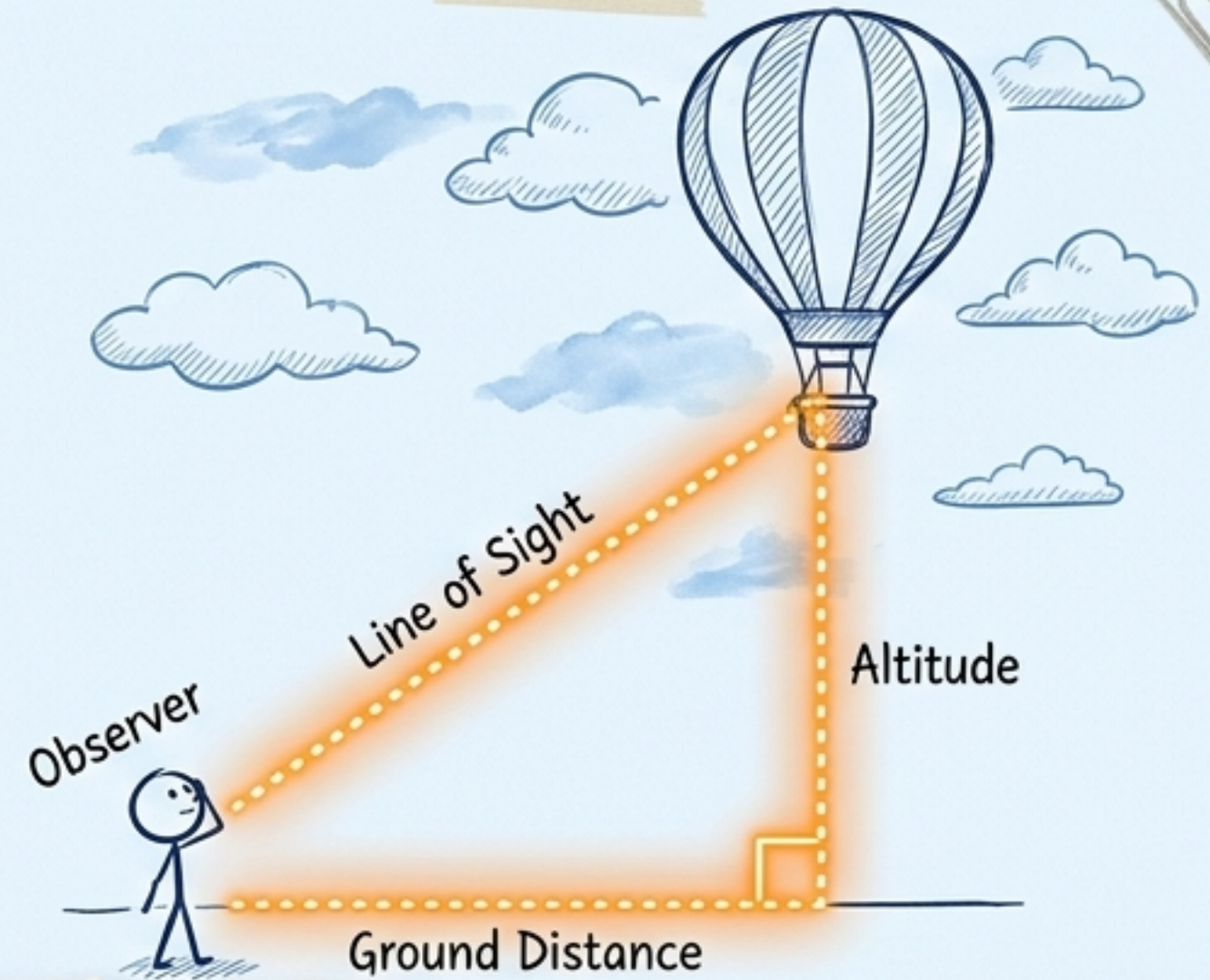
Everything you need to
survive Chapter 8.



**Save for
Exams!**



What even is Trigonometry?



Tri (3) + Gon (Sides) + Metron (Measure).
It's literally just the study of relationships
between the sides and angles of a triangle!

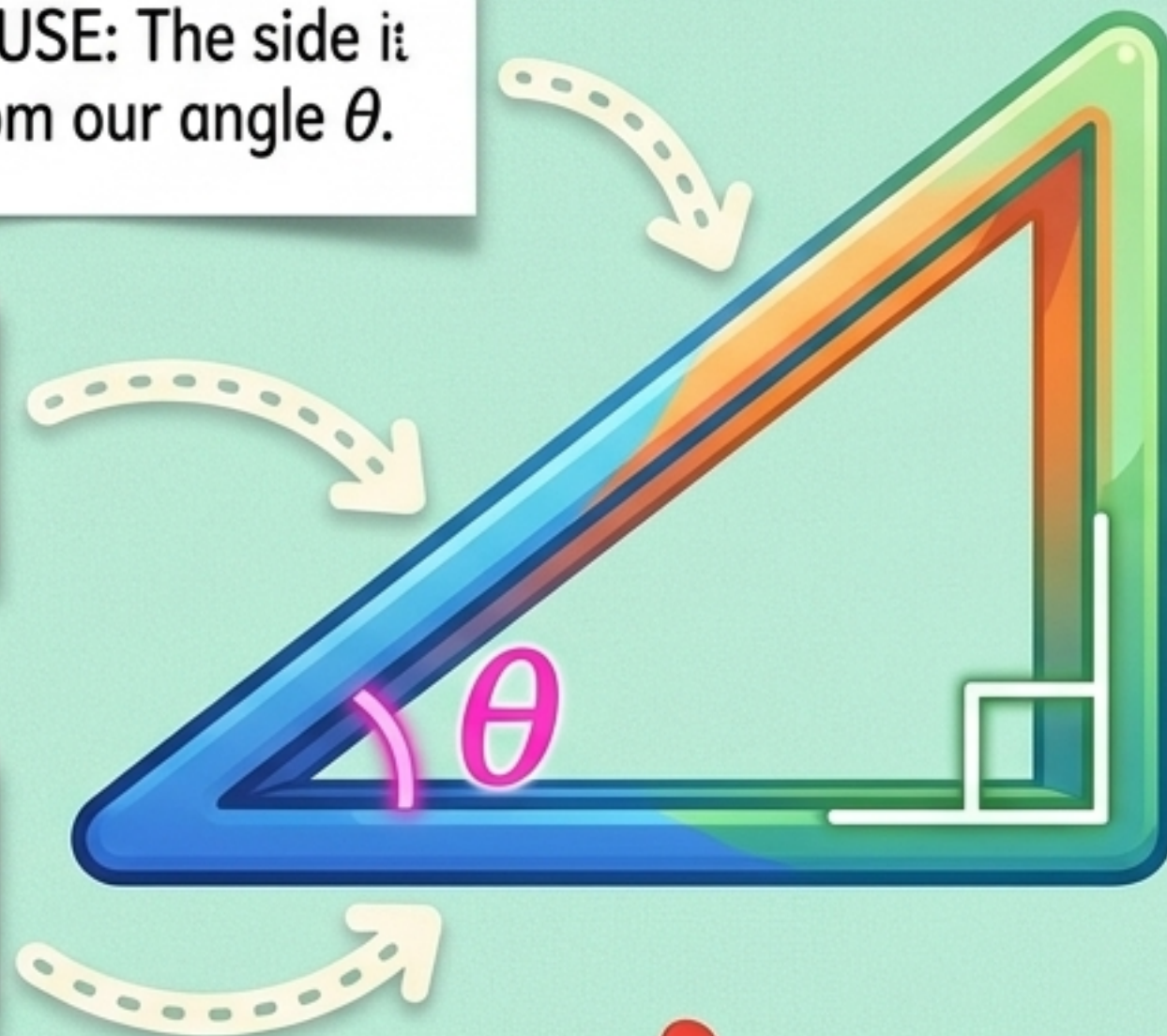


Anatomy of a Right Triangle

HYPOTENUSE: The side opposite to our angle θ .

OPPOSITE: The side opposite to our angle θ .

ADJACENT: The side adjacent to our angle θ .



HYPOTENUSE:
The longest side, always
across from the 90° angle.

⚠ Warning! "Opposite" and "Adjacent" change depending on which angle you are looking from!

The "Big 3" Ratios

SOH CAH TOA

SOH

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

CAH

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

TOA

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Pro-Tip:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Fun Fact: The word 'sine' comes from the Indian mathematician Aryabhata (500 CE) who called it 'ardha-jya' (half-chord)! 🤖

The Flip Side: Reciprocals 😊

$\sin A$

Just flip the
fraction
upside down!

**$\csc A$
(Cosecant)**

$\frac{\text{Hypotenuse}}{\text{Opposite}}$ (or $\frac{1}{\sin A}$)

$\cos A$

Just flip the
fraction
upside down!

**$\sec A$
(Secant)**

$\frac{\text{Hypotenuse}}{\text{Adjacent}}$ (or $\frac{1}{\cos A}$)

$\tan A$

Just flip the
fraction
upside down!

**$\cot A$
(Cotangent)**

$\frac{\text{Adjacent}}{\text{Opposite}}$ (or $\frac{1}{\tan A}$)

The Magic Angle Table ✨

	0°	30°	45°	60°	90°
$\sin A$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos A$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan A$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined



Secret Hack: The Sin/Cos X-Pattern 🤔

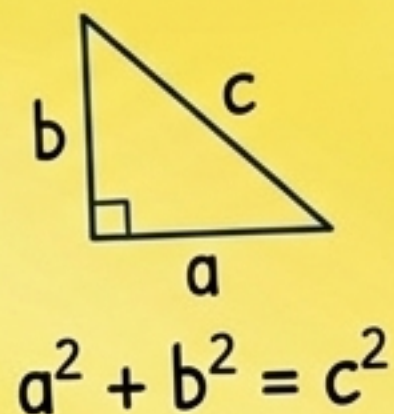
Angle	0°	30°	45°	60°	90°
sin A	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos A	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

Notice how Sine INCREASES from 0 to 1, while Cosine DECREASES from 1 to 0. They are exact reverse mirrors! 🤖 🤯

Trig Identities: The Master Keys


$$1 + \tan^2 A = \sec^2 A$$

(Divided by adjacent²)




$$\cos^2 A + \sin^2 A = 1$$

(Divided by hypotenuse²)


$$\cot^2 A + 1 = \csc^2 A$$

(Divided by opposite²)

Pro-Tip: Proving Identities



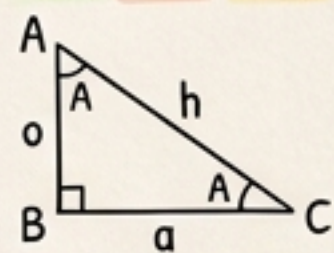
1. Pick a side.
Start with the more complicated/messy side (usually LHS).

3. Algebra Magic.
Look for common denominators,
factor out terms,
or use the identity
 $a^2 - b^2 = (a-b)(a+b)$.

2. The Golden Rule.
Convert EVERYTHING
to sin and cos.
(e.g., turn tan into sin/cos,
turn sec into 1/cos).



SOH CAH TOA



$$\cos^2 A + \sin^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$\cot^2 A + 1 = \csc^2 A$$

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	und

Tag a friend who needs to pass Math! 📌 ✨

